

EXPLORATORY DATA ANALYSIS

Experiment 1 – Basic Python Programs

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1. To calculate the area or perimeter of a rectangle

while True:

```
print("1. Area\n2. Perimeter\n3. Exit")
```

```
choice = int(input("Enter your choice: "))
```

```
if choice == 1:
```

```
    length = int(input("Enter the length of the rectangle: "))
```

```
    breadth = int(input("Enter the breadth of the rectangle: "))
```

```
    print("Area =", length * breadth)
```

```
elif choice == 2:
```

```
    length = int(input("Enter the length of the rectangle: "))
```

```
    breadth = int(input("Enter the breadth of the rectangle: "))
```

```
    print("Perimeter =", 2 * (length + breadth))
```

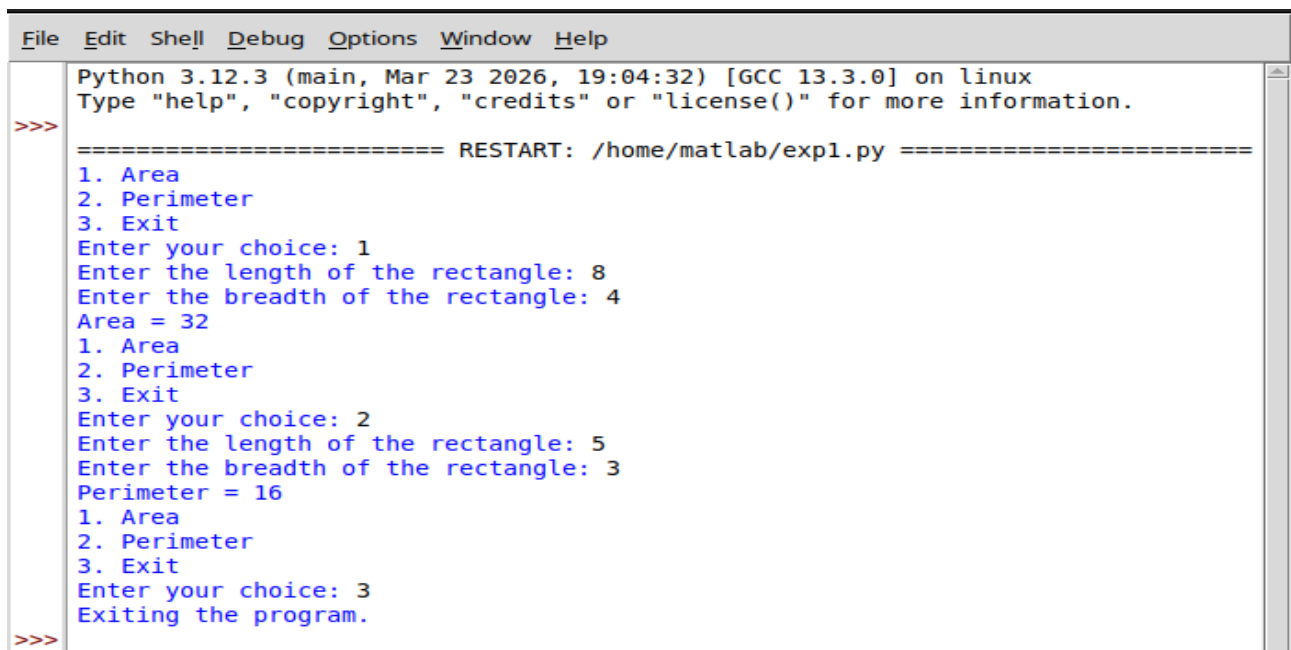
```
elif choice == 3:
```

```
    print("Exiting the program.")
```

```
    break
```

```
else:
```

```
    print("Invalid choice.")
```



The screenshot shows a terminal window with a menu-driven Python program. The program prompts the user to choose between calculating the area, perimeter, or exiting. In the first run, the user chooses '1' (Area), enters a length of 8 and a breadth of 4, and the program outputs 'Area = 32'. In the second run, the user chooses '2' (Perimeter), enters a length of 5 and a breadth of 3, and the program outputs 'Perimeter = 16'. In the third run, the user chooses '3' (Exit), and the program outputs 'Exiting the program.'.

```
File Edit Shell Debug Options Window Help
Python 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: /home/matlab/exp1.py =====
1. Area
2. Perimeter
3. Exit
Enter your choice: 1
Enter the length of the rectangle: 8
Enter the breadth of the rectangle: 4
Area = 32
1. Area
2. Perimeter
3. Exit
Enter your choice: 2
Enter the length of the rectangle: 5
Enter the breadth of the rectangle: 3
Perimeter = 16
1. Area
2. Perimeter
3. Exit
Enter your choice: 3
Exiting the program.
>>>
```

2. To check if a number is prime or not

```
num = int(input("Enter a number: "))
if num > 1:
    for i in range(2, num):
        if (num % i) == 0:
            print(num, "is not a prime number.")
            break
    else:
        print(num, "is a prime number.")
else:
    print(num, "is not a prime number.")
```

```
>>> ===== RESTART: /home/matlab/exp1.py =====
Enter a number: 3
3 is a prime number.
>>>
```

3. To check if a number is odd or even

```
num = int(input("Enter a number: "))
if num%2 == 0:
    print("It is an even number")
else:
    print("It is an odd number")
```

```
>>> ===== RESTART: /home/matlab/exp1.py =====
Enter a number: 2
It is an even number
>>>
```

4. Grade Calculator

```
marks = int(input("Enter your marks:"))
if(90<=marks<=100):
    print('A')
elif(80<=marks<90):
    print('B')
elif(70<=marks<80):
    print('C')
elif(60<=marks<70):
    print('D')
elif(50<=marks<60):
    print('E')
elif(marks<50):
    print('F')
```

```
>>> ===== RESTART: /home/matlab/exp1.py =====
Enter your marks:85
B
>>>
```

5. Armstrong Number

```
n = input("Enter a number: ")
num = int(n)
p = len(n)
total = 0
```

```
for d in n:
    total = total + (int(d) ** p)
```

```
if total == num:
    print(num, "is an Armstrong number.")
else:
    print(num, "is not an Armstrong number.")
```

```
===== RESTART: /home/matlab/exp1.py =====
Enter a number: 153
153 is an Armstrong number.
>>>
```

6. Palindrome Number

```
n = input("Enter a number: ")
rev = n[::-1]
```

```
if n == rev:
    print(n, "is a palindrome number.")
else:
    print(n, "is not a palindrome number.")
```

```
===== RESTART: /home/matlab/exp1.py =====
Enter a number: 111
111 is a palindrome number.
>>>
===== RESTART: /home/matlab/exp1.py =====
Enter a number: 123
123 is not a palindrome number.
>>>
```

7. To print the Fibonacci series

```
n = int(input("Enter the number of terms: "))
```

```
a = 0
b = 1
```

```
print("Fibonacci Series:")
```

```
for i in range(n):
    print(a, end=" ")
    c = a + b
    a = b
    b = c
```

```
>>> Type "help", "copyright", "credits" or "license()" for more information.
===== RESTART: C:/Users/suraj/Downloads/eda1.py =====
Enter the number of terms: 4
Fibonacci Series:
0 1 1 2
```

8. To reverse a string

```
string = input("Enter a string: ")
```

```
reverse = ""
```

```
for ch in string:
    reverse = ch + reverse
```

```
print("Reversed string:", reverse)
```

```
>>> ===== RESTART: C:/Users/suraj/Downloads/eda1.py =====
Enter a string: apple
Reversed string: elppa
>>>
```

9. To calculate the number of vowels and consonants

```
string = input("Enter a string: ").lower()
```

```
vowels = 0
```

```
consonants = 0
```

```
for ch in string:
    if ch.isalpha():
        if ch in "aeiou":
            vowels += 1
        else:
            consonants += 1
```

```
print("Vowels =", vowels)
```

```
print("Consonants =", consonants)
```

```
>>> ===== RESTART: C:/Users/suraj/Downloads/eda1.py =====
Enter a string: data
Vowels = 2
Consonants = 2
>>>
```

10. To find the largest of three numbers

```
n = int(input("Enter the number of elements: "))
```

```
arr = []
```

```
for i in range(n):
    num = int(input("Enter element: "))
    arr.append(num)
```

```
largest = arr[0]
```

```
for i in range(1, n):
```

```
if arr[i] > largest:  
    largest = arr[i]
```

```
print("Largest element =", largest)
```

```
==== RESTART: C:/Users/suraj/Downloads/eda1.py =====  
Enter the number of elements: 4  
Enter element: 1  
Enter element: 4  
Enter element: 5  
Enter element: 6  
Largest element = 6  
>>>
```